

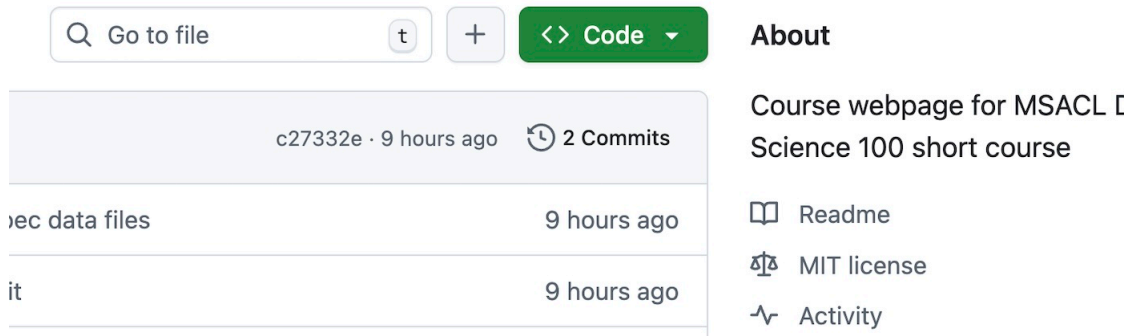
Lesson 1 Exercises

Exercises Setup

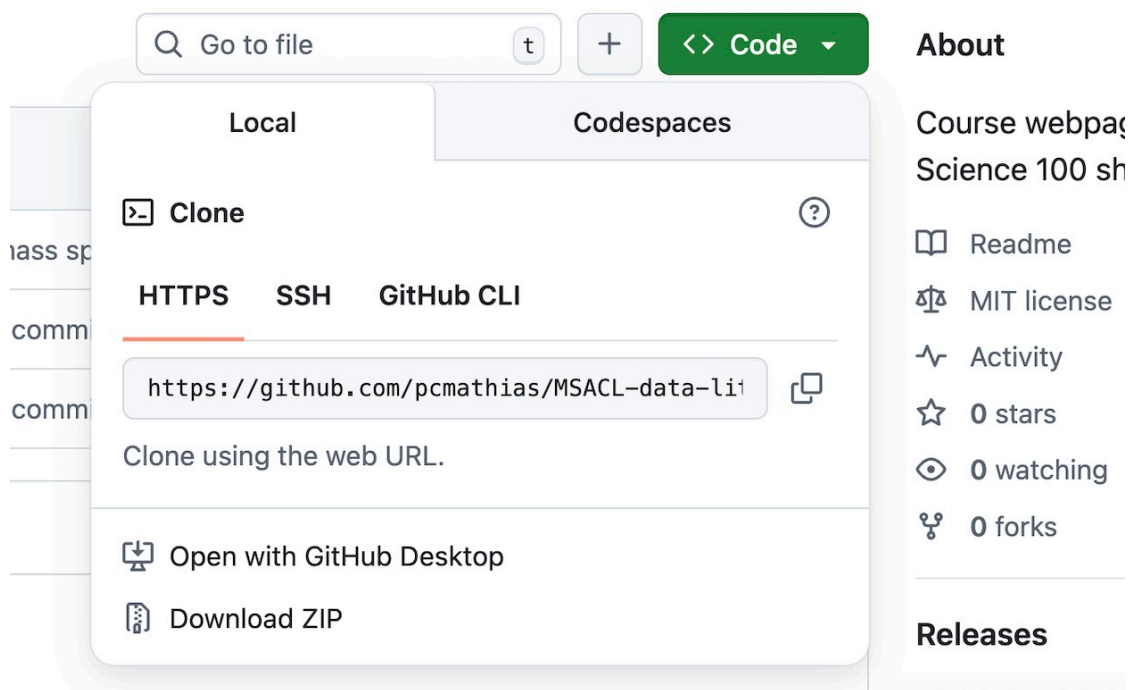
The materials for this course are hosted on a public Github website: <https://github.com/pcmathias/MSACL-data-literacy>. Github is a website that hosts repositories (repos), which are collections of files that are often organized into a project. You can think of each Github repository as a folder with files and subfolders. Github is a popular way to share code and projects with a broader community, who can contribute back to the repository. It also utilizes the git version control system, which allows users to track specific changes throughout the life cycle of a project.

To make accessing the course materials as easy as possible, we recommend downloading the repository for this course to your computer by following these steps.

1. Navigate to the course website in your browser
2. Press the “Code” button near the top right of the page



3. Click “Download ZIP” to download a .zip file that contains the repository



4. Place the .zip file in a location you will remember on your computer and unzip it. The best practice is to create a “Projects” folder in which you create a new folder for each project.

Orientation to the Data Set

Throughout this short course we will refer to a mass spectrometry data set that includes three files. The batch data file includes data that capture information about six analytes that are measured on 20 batches per day over the course of a month. The sample data file includes information about each sample in each batch, which is composed of 52 samples. Each sample has 24 peaks with data on each peak captured in the peak data file.

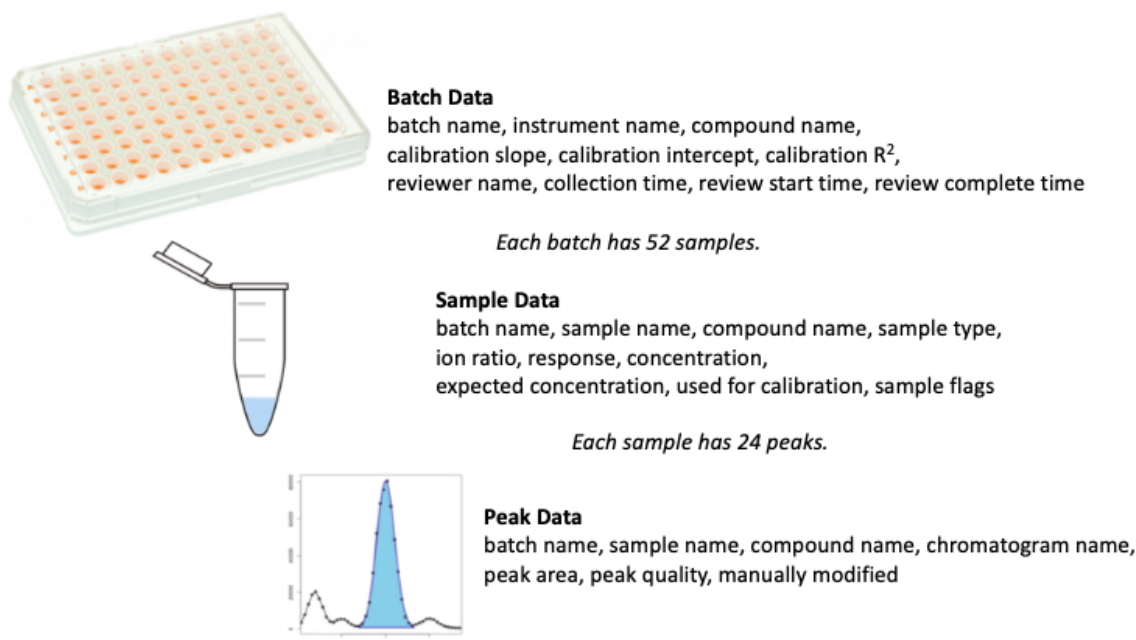


Figure 1: Summary of mass spec data set

Exercise 1: Data Types

We will explore data types in our mass spec data set and how different approaches to interacting with the data set may provide different views or perspectives of the data.

1. In the *exercises* folder within the main project folder, open the **Lesson 1 Exercise 1 Worksheet.xlsx** file.
2. Navigate to the *data* folder within the main project folder. Right click the **batch_data.csv** file and open the file with a text application such as Notepad for Windows or TextEdit for Macs.
3. Right click the **batch_data.csv** file and open with Excel. (Note that there is a similarly named **batch_data_ts.csv** file in the data folder as well. Do not open this yet.) If there is a pop-up message that prompts you to take action, read the message carefully and click “Convert.”
4. For each of the column names (separated by commas in the text application), complete the **Lesson 1 Exercise 1 Worksheet.xlsx** based on the data types that you manually infer based on what we just learned about data types from the text file, and based on the format window in Excel.

5. Working the `batch_data.csv` file, click the File menu at the top left and click “Save As...” The file format to save should be defaulted to “Comma Separated Values (.csv)” - if it is not, select that and update the file name to “`batch_data_excel.csv`.” Open the **`batch_data_excel.csv`** file using your text application. How is it different from the original **`batch_data.csv`** file that you previously opened within the text application?
6. Open the **`batch_data_ts.csv`** file in Excel. What is different in this file compared to the previous file?
7. Use Save As... to save a new file. Create a new folder called “output” within the MSACL-data-literacy folder. Select the new folder. Save the new file as an Excel (.xlsx) file named **`batch_data_excel.xlsx`**.
8. Click on cell M2. In the formula box, enter the following:

“=DATEVALUE(MID(I2,1,10))+TIMEVALUE(MID(I2,12,8))”

The MID function extracts a specified number of characters from a text string, starting at a certain position. The formula will take the first 10 characters of the timestamp text and convert it into a date as a floating point value, take the 8 characters from the 12th position from the left and convert it into a time as a floating point value, and then sum the numeric values together.

9. Convert the floating point number to a date and time recognized by Excel by right-clicking the cell, clicking on “Format Cells...”, selecting “Time” from the Category list, and selecting the bottom most option in the Type window (3/14/12 13:30).

Exercise 2: Tidy Data

Many, but not all, systems and workflows will generate tidy data, where each variable is in its own column, each observation is in its own row, and each value is in its own cell. Transactional data that captures specific events and timestamps often is organized in this format, but sometimes the software for an instrument does not export data in this format.

1. Open **`sample_data_wide.csv`** in Excel. Does this look like tidy data? If not, which of the 3 principles of tidy data does it violate?
2. Navigate to the Data menu in Excel and click the down arrow beside the Get Data button.

AutoSave ☐ Home Insert Draw Page Layout Formulas **Data** Review View Automate Developer Acrobat

Get Data (Power Query) Discover, connect, and combine data from multiple sources

Stocks Currencies Geography

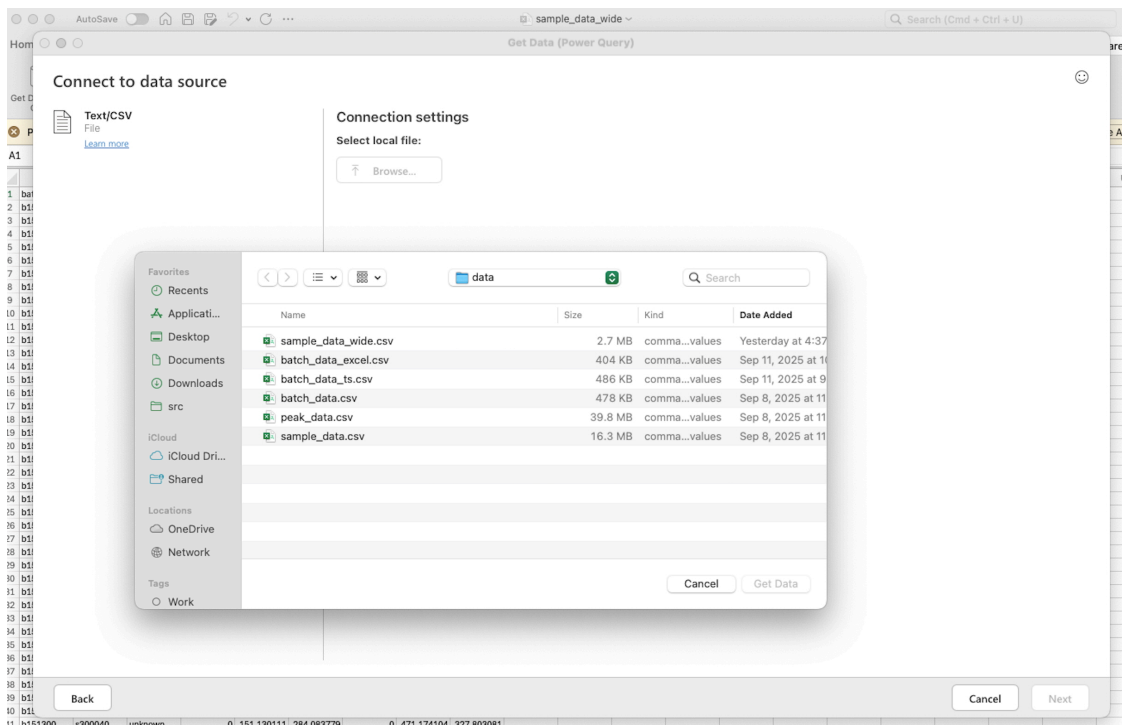
Sort Filter Reapply Advanced

⚠ Possible Data Loss Some features might be lost if you save this workbook in the comma-delimited (.csv) format. To preserve these features, save it in

A1

	A	B	C	D	E	F	G	H	I	J	K	L	M
	batch_name	sample_name	sample_type	morphine	hydromorphone	oxycodone	codeine	hydrocodone	oxycodone				
1	batch_name	sample_name	sample_type	morphine	hydromorphone	oxycodone	codeine	hydrocodone	oxycodone				
2	b151300	s300001	blank	0	0	0	0	0	0				
3	b151300	s300002	standard	0	0	0	0	0	0				
4	b151300	s300003	standard	18.2650536	22.2802585	22.5361657	21.3820948	17.5340655	20.4006466				
5	b151300	s300004	standard	46.7722649	48.3041454	49.0685495	46.3681972	59.3187673	44.2984216				
6	b151300	s300005	standard	105.56442	93.5477538	87.1492615	108.080928	89.0321093	100.78411				
7	b151300	s300006	standard	246.632568	165.785991	189.425014	179.056843	190.49174	202.29274				
8	b151300	s300007	standard	256.648993	330.791846	323.224129	295.844364	318.454301	330.647732				
9	b151300	s300008	standard	503.49917	526.428596	497.89464	520.64497	516.313624	486.267667				
10	b151300	s300009	blank	0	0	0	0	0	0				
11	b151300	s300010	unknown	0	401.050093	147.892546	0	196.450891	476.638706				

3. In the Choose data source menu select Text/CSV.
4. Click the Browse button and select **sample_data_wide.csv**, then click the Get Data button.



5. Hit the Next button and you should see your data set on the next page. Excel provides some information about the file and contents above and shows an indicator of the data types for each column by the column name (text vs. numeric).

The screenshot displays the Power Query Editor window. The main area shows a table with the following columns: `sample_name`, `sample_type`, `1,2 morphine`, `1,2 hydromorphone`, `1,2 oxycodone`, `1,2 codeine`, `1,2 hydrocodone`, and `1,2 oxycodone`. The table contains 27 rows of data. The right-hand pane shows the 'Query settings' for 'sample_data_wide', including 'Properties' and 'Applied steps'.

sample_name	sample_type	1,2 morphine	1,2 hydromorphone	1,2 oxycodone	1,2 codeine	1,2 hydrocodone	1,2 oxycodone
1	blank	0	0	0	0	0	0
2	standard	0	0	0	0	0	0
3	standard	18.26505358	22.28025847	22.53616566	21.38209482	17.53406553	20.40064659
4	standard	46.77226485	48.30414539	49.06854948	46.36819723	59.31876731	44.29842158
5	standard	105.5644201	93.54775382	87.14926152	108.0809277	89.03210932	100.7841096
6	standard	248.6325682	165.7859908	189.4250137	179.0568434	190.4917403	202.2927404
7	standard	256.6489926	330.7918455	323.2241293	295.8443642	318.4543005	330.6477322
8	standard	503.4991702	526.4285962	497.8946397	520.64497	516.3136236	486.2676667
9	blank	0	0	0	0	0	0
10	unknown	0	401.0500927	147.8925455	0	196.4508909	476.6387061
11	unknown	0	335.1722412	134.9415646	0	0	0
12	unknown	0	270.8745166	0	0	482.992252	0
13	unknown	0	0	255.7861615	0	0	0
14	unknown	430.7035783	160.5329416	0	385.6338339	0	367.094119
15	unknown	0	45.89784272	0	0	267.7488804	0
16	unknown	488.6318384	429.1436357	240.7405429	0	388.1793845	123.7500762
17	unknown	0	0	0	0	0	452.4343497
18	unknown	252.6245894	0	0	0	269.8762986	177.7428869
19	unknown	0	493.5033308	140.622889	0	427.9886787	0
20	qc	37.66722344	42.22958168	31.58089887	36.19921501	42.62637347	33.94849425
21	unknown	149.5925696	0	203.4154292	0	0	0
22	unknown	259.6329715	161.9099871	0	274.2058262	359.1532878	48.26142788
23	unknown	471.7436989	0	0	0	145.6775285	82.47047734
24	unknown	0	37.12346531	254.7889628	0	0	0
25	unknown	0	395.653517	92.15582903	38.4914365	330.3137079	0
26	unknown	222.6574912	464.1591963	202.924407	0	0	224.1337572
27							

- Hit the Transform data button to bring up a different menu. You are now within the Power Query Editor, which allows for some transformations.

Power Query Editor

Table.TransformColumnTypes({"Promoted headers", {"batch_name", type text}, {"sample_name", type text}, {"sample_type", type text}, {"1,2 morphine", type decimal}, {"1,2 hydromorphone", type decimal}, {"1,2 oxymorphone", type decimal}, {"1,2 codeine", type decimal}, {"1,2 hydrocodone", type decimal}, {"1,2 oxycodone", type decimal}})

sample_name	sample_type	1,2 morphine	1,2 hydromorphone	1,2 oxymorphone	1,2 codeine	1,2 hydrocodone	1,2 oxycodone
1	blank	0	0	0	0	0	0
2	standard	0	0	0	0	0	0
3	standard	18.26505358	22.28025847	22.53616566	21.38209482	17.53406553	20.40064659
4	standard	46.77226485	48.30414539	49.06854948	46.36819723	59.31876731	44.29842158
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6	standard	246.6325682	165.7859906	189.4250137	179.0568434	190.4917403	202.2927404
7	standard	256.6489926	330.7918455	323.2241293	295.8443642	318.4543005	330.6477222
8	standard	503.4991702	526.4285962	497.8946397	520.64497	516.3136236	486.2676667
9	blank	0	0	0	0	0	0
10	unknown	0	401.0500927	147.8925455	0	196.4508909	476.6387061
11	unknown	0	335.1722412	134.9415646	0	0	0
12	unknown	0	270.8745166	0	0	482.992252	0
13	unknown	0	0	255.7861615	0	0	0
14	unknown	430.7035783	160.5329416	0	385.6338339	0	367.094119
15	unknown	0	45.89784272	0	0	267.7488804	0
16	unknown	488.6318384	429.1436357	240.7405429	0	388.1793845	123.7500762
17	unknown	0	0	0	0	0	452.4343497
18	unknown	252.6245894	0	0	0	269.8762986	177.7428869
19	unknown	0	493.5033308	140.622689	0	427.9886787	0
20	qc	37.66722344	42.22958168	31.58089887	36.19921501	42.62637347	33.94849425
21	unknown	149.5925696	0	203.4154292	0	0	0
22	unknown	259.6329715	161.9099871	0	274.2058262	359.1532878	48.26142788
23	unknown	471.7436989	0	0	0	145.6775285	82.47047734
24	unknown	0	37.12346531	254.7889628	0	0	0
25	unknown	0	395.653517	92.15582903	38.4914365	330.3137079	0
26	unknown	222.6574912	464.1591963	202.924407	0	0	224.1337572
27	.	0	0	0	0	0	0

Query settings

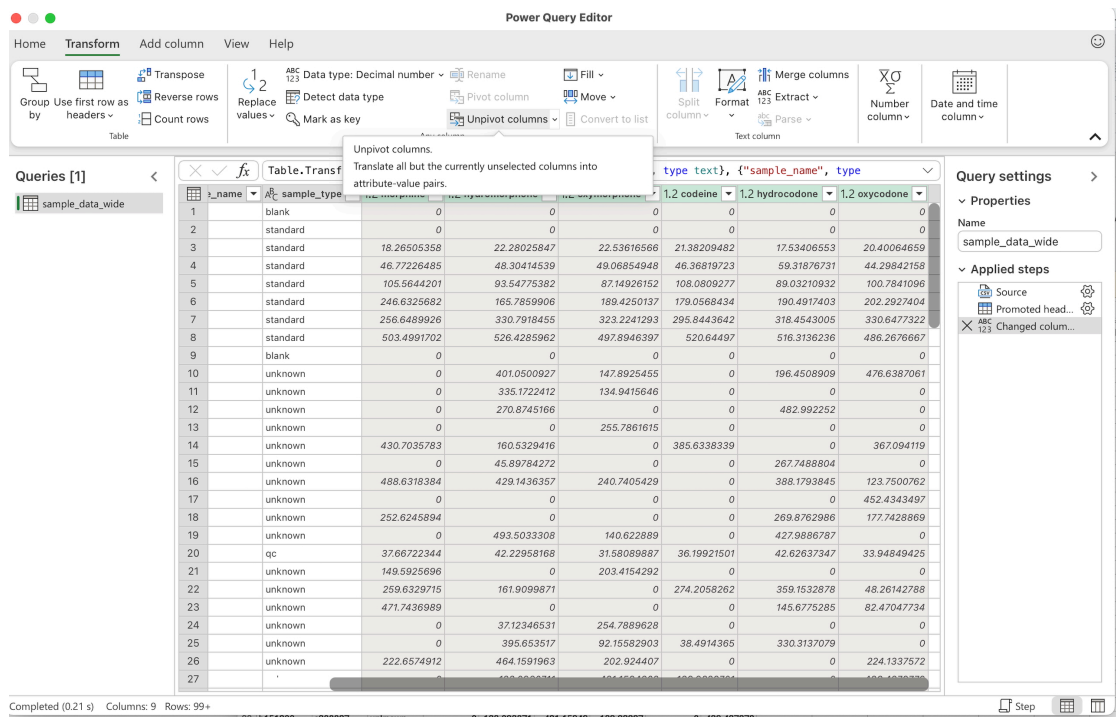
Properties

Name: sample_data_wide

Applied steps

- Source
- Promoted head...
- Changed column...

- Click the morphine column, hold the shift button, and then click the final oxycodone column. All 6 analyte columns should be highlighted.
- Click the Unpivot columns button. The data in the window should change and you should now see one concentration value for a single analyte per row.



9. Navigate to the Home menu and hit the Close & load button to load the data into an Excel worksheet. Note how long this takes to complete.

Data Transformation in R

R and other programming languages such as Python that are intended to transform data are efficient at these types of operations. The equivalent operations of loading and viewing the data can be performed with the following code:

```
library(tidyverse)
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
v dplyr      1.1.4      v readr      2.1.4
v forcats    1.0.0      v stringr    1.5.1
v ggplot2    3.5.1      v tibble     3.2.1
v lubridate  1.9.2      v tidyr      1.3.1
v purrr      1.0.2

-- Conflicts ----- tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()     masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```



```
# loads "packages" that allow the user to use functions that others have written
sample_wide <- read_csv("../data/sample_data_wide.csv")
```

```
Rows: 31148 Columns: 9
```

```
-- Column specification -----
```

```
Delimiter: ","
```

```
chr (3): batch_name, sample_name, sample_type
```

```
dbl (6): morphine, hydromorphone, oxymorphone, codeine, hydrocodone, oxycodone
```

```
i Use `spec()` to retrieve the full column specification for this data.
```

```
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
# reads the sample_data_wide.csv file into R
```

```
head(sample_wide, 10) # displays the first 10 rows
```

```
# A tibble: 10 x 9
```

	batch_name	sample_name	sample_type	morphine	hydromorphone	oxymorphone	codeine
	<chr>	<chr>	<chr>	<dbl>	<dbl>	<dbl>	<dbl>
1	b151300	s300001	blank	0	0	0	0
2	b151300	s300002	standard	0	0	0	0
3	b151300	s300003	standard	18.3	22.3	22.5	21.4
4	b151300	s300004	standard	46.8	48.3	49.1	46.4
5	b151300	s300005	standard	106.	93.5	87.1	108.
6	b151300	s300006	standard	247.	166.	189.	179.
7	b151300	s300007	standard	257.	331.	323.	296.
8	b151300	s300008	standard	503.	526.	498.	521.
9	b151300	s300009	blank	0	0	0	0
10	b151300	s300010	unknown	0	401.	148.	0

```
# i 2 more variables: hydrocodone <dbl>, oxycodone <dbl>
```

Data transformation into a tidy data set requires only one function:

```
sample <- pivot_longer(sample_wide,
                        cols = morphine:oxycodone,
                        names_to = "analyte",
                        values_to = "concentration")
head(sample, 10) # view the first 10 lines of the new dataframe
```

```
# A tibble: 10 x 5
```

batch_name	sample_name	sample_type	analyte	concentration
------------	-------------	-------------	---------	---------------

	<chr>	<chr>	<chr>	<chr>	<dbl>
1	b151300	s300001	blank	morphine	0
2	b151300	s300001	blank	hydromorphone	0
3	b151300	s300001	blank	oxymorphone	0
4	b151300	s300001	blank	codeine	0
5	b151300	s300001	blank	hydrocodone	0
6	b151300	s300001	blank	oxycodone	0
7	b151300	s300002	standard	morphine	0
8	b151300	s300002	standard	hydromorphone	0
9	b151300	s300002	standard	oxymorphone	0
10	b151300	s300002	standard	codeine	0